Lecture 9: Large-sample properties of OLS and robust standard errors

Economics 527: Econometric Methods

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Consistency of OLS

OLS in terms of sample averages

- Regression, $\{(Y_i, X_i) : i = 1, ..., n\}$ iid:

$$Y_i = X_i^\top \beta + U_i, \qquad X_i \in \mathbb{R}^k$$

- OLS, $\text{rank}(X) = k$:

$$\begin{aligned}\hat{\beta}_n &= (X^\top X)^{-1} X^\top Y \\ &= \beta + \Big(\frac{1}{n}\sum_{i=1}^n X_i X_i^\top\Big)^{-1} \frac{1}{n}\sum_{i=1}^n X_i U_i\end{aligned}$$

LLN for $\frac{1}{n}\sum X_i U_i$

- **Assume:**

$$\mathrm{E}[X_i U_i] = 0, \qquad \mathrm{E}[U_i^2] < \infty, \qquad \mathrm{E}\left[X_{i,j}^2\right] < \infty, \quad j = 1, ..., k$$

- $X_1 U_1, ..., X_n U_n$: iid; weak LLN, element by element:

$$\frac{1}{n}\sum_{i=1}^n X_i U_i \to_p \mathrm{E}[X_i U_i] = 0$$

LLN for $\frac{1}{n}\sum X_i X_i^\top$

- Cauchy-Schwarz, element (r, s):

$$\mathrm{E}\left[|X_{i,r} X_{i,s}|\right] \le \sqrt{\mathrm{E}\left[X_{i,r}^2\right] \mathrm{E}\left[X_{i,s}^2\right]} < \infty$$

- **Assume:** $\mathrm{E}[X_i X_i^\top]$ p.d.
- Weak LLN and Slutsky for the inverse:

$$\frac{1}{n}\sum_{i=1}^n X_i X_i^\top \to_p \mathrm{E}[X_i X_i^\top]$$

$$\Big(\frac{1}{n}\sum_{i=1}^n X_i X_i^\top\Big)^{-1} \to_p \left(\mathrm{E}[X_i X_i^\top]\right)^{-1}$$

Assumptions

- Assumptions for consistency:

$$\{(Y_i, X_i) : i = 1, \dots, n\} \quad \text{iid}$$
$$Y_i = X_i^\top \beta + U_i$$
$$\mathrm{E}\left[X_i U_i\right] = 0$$
$$\mathrm{E}\left[U_i^2\right] < \infty$$
$$\mathrm{E}\left[X_{i,j}^2\right] < \infty, \quad j = 1, \dots, k$$
$$\mathrm{E}\left[X_i X_i^\top\right] \quad \text{p.d. (no perfect multicollinearity)}$$

- Weaker than $\mathrm{E}\left[U_i \mid X_i\right] = 0$:

$$\mathrm{E}\left[X_i U_i\right] = 0 \quad \not\Rightarrow \quad \mathrm{E}\left[Y_i \mid X_i\right] = X_i^\top \beta \quad \text{(in general)}$$

Consistency

- **Definition.** $\hat{\beta}_n$ consistent for β: $\hat{\beta}_n \to_p \beta$.
- **Theorem.** Under the assumptions for consistency:

$$\hat{\beta}_n = \beta + \Big(\underbrace{\frac{1}{n} \sum_{i=1}^n X_i X_i^\top}_{\to_p \mathrm{E}[X_i X_i^\top]} \Big)^{-1} \underbrace{\frac{1}{n} \sum_{i=1}^n X_i U_i}_{\to_p 0}$$
$$\to_p \beta \qquad \text{(Slutsky)}$$

Asymptotic normality of OLS

Assumptions I and II

- Approximate the distribution of $\hat{\beta}_n - \beta$: scale by $\sqrt{n}$.
- **Assumption I.**

$$\{(Y_i, X_i) : i = 1, \dots, n\} \quad \text{iid}$$
$$Y_i = X_i^\top \beta + U_i, \qquad \beta \in \mathbb{R}^k$$

- **Assumption II.**

$$Q = \mathrm{E}\left[X_i X_i^\top\right] \quad \text{p.d. and finite}$$

- Normalized OLS representation:

$$\sqrt{n}(\hat{\beta}_n - \beta) = \underbrace{\Big(\frac{1}{n} \sum_{i=1}^n X_i X_i^\top \Big)^{-1}}_{\to_p Q^{-1}} \frac{1}{\sqrt{n}} \sum_{i=1}^n X_i U_i$$

Assumption III and the CLT

- **Assumption III.**

$$\mathrm{E}\left[X_i U_i\right] = 0, \qquad \Omega = \mathrm{Var}\left(X_i U_i\right) \quad \text{finite and p.d.}$$

- Variance of $X_i U_i$, with $\mathrm{E}\left[X_i U_i\right] = 0$:

$$\Omega = \mathrm{Var}\left(X_i U_i\right) = \mathrm{E}\left[X_i U_i (X_i U_i)^\top\right] = \mathrm{E}\left[U_i^2 X_i X_i^\top\right]$$

- $X_1 U_1, \dots, X_n U_n$: iid; multivariate CLT:

$$\frac{1}{\sqrt{n}} \sum_{i=1}^n X_i U_i \to_d N(0, \Omega)$$

Limit of $\sqrt{n}(\hat{\beta}_n - \beta)$

- Cramér convergence theorem, without proof: $A_n \to_p A$, A non-random, $Z_n \to_d Z$:

$$A_n Z_n \to_d AZ$$

- Apply to the normalized representation; $Z \sim N(0, \Omega)$:

$$\begin{aligned}\sqrt{n}(\hat{\beta}_n - \beta) &= \underbrace{\Big(\frac{1}{n}\sum_{i=1}^{n} X_i X_i^\top\Big)^{-1}}_{\to_p Q^{-1}} \underbrace{\frac{1}{\sqrt{n}}\sum_{i=1}^{n} X_i U_i}_{\to_d Z} \\ &\to_d Q^{-1} Z \sim N(0, Q^{-1}\Omega Q^{-1}) \qquad (Q = Q^\top)\end{aligned}$$

Asymptotic variance

- **Theorem.** Under Assumptions I to III:

$$\sqrt{n}(\hat{\beta}_n - \beta) \to_d N(0, V), \qquad V = \text{AsyVar}(\hat{\beta}_n) = Q^{-1}\Omega Q^{-1}$$

- CDF form, for all $a \neq 0$ and $t \in \mathbb{R}$:

$$\text{P}(a^\top \sqrt{n}(\hat{\beta}_n - \beta) \leq t) \to \Phi\Big(\frac{t}{\sqrt{a^\top V a}}\Big)$$

- $a = e_j$, element j:

$$\sqrt{n}(\hat{\beta}_{n,j} - \beta_j) \to_d N(0, [V]_{jj})$$

Homoskedasticity

- Suppose:

$$\text{E}\left[U_i^2 \mid X_i\right] = \sigma^2 \quad \text{(homoskedasticity)}$$

- Law of iterated expectations:

$$\begin{aligned}\Omega &= \text{E}\left[U_i^2 X_i X_i^\top\right] \\ &= \text{E}\left[\text{E}\left[U_i^2 X_i X_i^\top \mid X_i\right]\right] \\ &= \text{E}\left[\underbrace{\text{E}\left[U_i^2 \mid X_i\right]}_{=\sigma^2} X_i X_i^\top\right] \\ &= \sigma^2 \text{E}\left[X_i X_i^\top\right] \\ &= \sigma^2 Q\end{aligned}$$

- The sandwich collapses:

$$\begin{aligned}\text{AsyVar}(\hat{\beta}_n) &= Q^{-1}\Omega Q^{-1} \\ &= Q^{-1}(\sigma^2 Q)Q^{-1} \\ &= \sigma^2 Q^{-1}\end{aligned}$$

Finite sample against large sample

- Normal errors, exact conditional distribution; $a \neq 0$:

$$\begin{aligned}& Y = X\beta + U, \qquad U \mid X \sim N(0, \sigma^2 I_n) \\ \implies\ & \hat{\beta}_n \mid X \sim N(\beta,\ \sigma^2 (X^\top X)^{-1}) \\ \implies\ & \text{P}(a^\top(\hat{\beta}_n - \beta) \leq t \mid X) = \Phi\Big(\frac{t}{\sqrt{\sigma^2 a^\top (X^\top X)^{-1} a}}\Big)\end{aligned}$$

- Homoskedastic errors, large sample, $a \neq 0$:

$$\begin{aligned} & \text{Assumptions I and II,} \quad \mathrm{E}\left[X_i U_i\right] = 0, \quad \mathrm{E}\left[U_i^2 \mid X_i\right] = \sigma^2 > 0 \\ \implies \quad & \mathrm{P}(a^\top \sqrt{n}(\hat{\beta}_n - \beta) \leq t) \to \Phi\Big(\frac{t}{\sqrt{\sigma^2 a^\top (\mathrm{E}\left[X_i X_i^\top\right])^{-1} a}}\Big) \end{aligned}$$

Estimation of the asymptotic variance

The estimator $\hat{V}_n$

- Unknown asymptotic variance:

$$V = Q^{-1}\Omega Q^{-1}, \qquad Q = \mathrm{E}\left[X_i X_i^\top\right], \qquad \Omega = \mathrm{E}\left[U_i^2 X_i X_i^\top\right]$$

- Sample averages and residuals in place of expectations and errors:

$$\begin{aligned} \hat{V}_n &= \hat{Q}_n^{-1} \hat{\Omega}_n \hat{Q}_n^{-1}, \\ \hat{Q}_n &= \frac{1}{n}\sum_{i=1}^n X_i X_i^\top, \qquad \hat{Q}_n^{-1} \to_p Q^{-1}, \\ \hat{\Omega}_n &= \frac{1}{n}\sum_{i=1}^n \hat{U}_i^2 X_i X_i^\top \end{aligned}$$

- Residual and error, fixed i; continuous mapping, without proof:

$$\begin{aligned} \hat{U}_i &= Y_i - X_i^\top \hat{\beta}_n = U_i + X_i^\top \underbrace{(\beta - \hat{\beta}_n)}_{\to_p 0} \\ &\to_p U_i \end{aligned}$$

Can we use the LLN on $\hat{\Omega}_n$?

- **Definition.** Small o_p:

$$Z_n = o_p(1) \quad \iff \quad Z_n \to_p 0$$

- Fixed i: $\hat{U}_i = U_i + o_p^i(1)$:

$$\hat{\Omega}_n = \frac{1}{n}\sum_{i=1}^n \left(U_i + o_p^i(1)\right)^2 X_i X_i^\top \stackrel{?}{=} \frac{1}{n}\sum_{i=1}^n U_i^2 X_i X_i^\top + o_p(1)$$

- Fixed-i convergence does not control an average over n observations.
- Direct iid LLN unavailable: residuals share $\hat{\beta}_n$:

$$\hat{U}_i = U_i - X_i^\top (\hat{\beta}_n - \beta)$$

Matrix norm

- Frobenius norm of a $k \times k$ matrix A:

$$\|A\| = \sqrt{\sum_{r=1}^k \sum_{s=1}^k A_{rs}^2} = \sqrt{\mathrm{tr}(A^\top A)}$$

- Element by element $\iff$ in norm; to show:

$$\|\hat{\Omega}_n - \Omega\| \to_p 0$$

Expansion of $\hat{\Omega}_n$

- Substitute $\hat{U}_i = U_i - X_i^\top(\hat{\beta}_n - \beta)$:

$$\begin{aligned}\hat{\Omega}_n &= \frac{1}{n}\sum_{i=1}^n \left(U_i - X_i^\top(\hat{\beta}_n - \beta)\right)^2 X_i X_i^\top \\ &= \frac{1}{n}\sum_{i=1}^n U_i^2 X_i X_i^\top + \frac{1}{n}\sum_{i=1}^n \left(X_i^\top(\hat{\beta}_n - \beta)\right)^2 X_i X_i^\top \\ &\quad - 2\,\frac{1}{n}\sum_{i=1}^n \left(U_i X_i^\top(\hat{\beta}_n - \beta)\right) X_i X_i^\top\end{aligned}$$

- First term, weak LLN; Ω finite (Assumption III):

$$\frac{1}{n}\sum_{i=1}^n U_i^2 X_i X_i^\top \to_p \Omega$$

- To show: the other two terms $\to_p 0$.

Cauchy-Schwarz and $\|X_i X_i^\top\|$

- Cauchy-Schwarz:

$$\left|X_i^\top(\hat{\beta}_n - \beta)\right| \le \|X_i\|\,\|\hat{\beta}_n - \beta\|$$

- Frobenius norm, $\operatorname{tr}(AB) = \operatorname{tr}(BA)$:

$$\begin{aligned}\|X_i X_i^\top\| &= \sqrt{\operatorname{tr}(X_i X_i^\top X_i X_i^\top)} \\ &= \sqrt{(X_i^\top X_i)^2} = \|X_i\|^2\end{aligned}$$

Cross term $\to_p 0$

- **Assume:** $\mathrm{E}\left[U_i^4\right] < \infty$, $\mathrm{E}\left[\|X_i\|^4\right] < \infty$.
- Cauchy-Schwarz twice:

$$\begin{aligned}\mathrm{E}\left[|U_i|\,\|X_i\|^3\right] &\le \sqrt{\mathrm{E}\left[U_i^2\|X_i\|^2\right]\,\mathrm{E}\left[\|X_i\|^4\right]} \\ &\le \sqrt{\sqrt{\mathrm{E}\left[U_i^4\right]\,\mathrm{E}\left[\|X_i\|^4\right]}\cdot \mathrm{E}\left[\|X_i\|^4\right]} < \infty\end{aligned}$$

- Triangle inequality, Cauchy-Schwarz, then weak LLN and consistency:

$$\begin{aligned}\left\|\frac{1}{n}\sum_{i=1}^n \left(U_i X_i^\top(\hat{\beta}_n - \beta)\right) X_i X_i^\top\right\| &\le \frac{1}{n}\sum_{i=1}^n |U_i|\left|X_i^\top(\hat{\beta}_n - \beta)\right|\|X_i X_i^\top\| \\ &\le \Big(\underbrace{\frac{1}{n}\sum_{i=1}^n |U_i|\,\|X_i\|^3}_{\to_p \mathrm{E}[|U_i|\,\|X_i\|^3]}\Big)\,\underbrace{\|\hat{\beta}_n - \beta\|}_{\to_p 0} \\ &\to_p 0\end{aligned}$$

Squared term $\to_p 0$

- Triangle inequality and Cauchy-Schwarz, then weak LLN and consistency:

$$\begin{aligned}\Big\|\frac{1}{n}\sum_{i=1}^n \big(X_i^\top(\hat\beta_n-\beta)\big)^2 X_iX_i^\top\Big\| &\le \frac{1}{n}\sum_{i=1}^n \big(X_i^\top(\hat\beta_n-\beta)\big)^2 \|X_iX_i^\top\| \\ &\le \Big(\underbrace{\frac{1}{n}\sum_{i=1}^n \|X_i\|^4}_{\to_p \mathrm{E}[\|X_i\|^4]<\infty}\Big)\underbrace{\|\hat\beta_n-\beta\|^2}_{\to_p 0} \\ &\to_p 0\end{aligned}$$

$\hat\Omega_n \to_p \Omega$ and $\hat V_n \to_p V$

- Combine the three terms:

$$\hat\Omega_n = \underbrace{\frac{1}{n}\sum_{i=1}^n U_i^2 X_iX_i^\top}_{\to_p \Omega} + o_p(1) \to_p \Omega$$

- **Theorem.** Assumptions I to III, $\mathrm{E}\left[U_i^4\right]<\infty$, $\mathrm{E}\left[\|X_i\|^4\right]<\infty$:

$$\hat V_n = \hat Q_n^{-1}\hat\Omega_n\hat Q_n^{-1} \to_p Q^{-1}\Omega Q^{-1} = V \qquad \text{(Slutsky)}$$

Robust and usual standard errors

Robust standard errors

- $\overset{a}{\sim}$: approximate distribution for large n:

$$\hat\beta_{n,j} \overset{a}{\sim} N\Big(\beta_j,\ \frac{[V]_{jj}}{n}\Big)$$

- **Definition.** Robust (heteroskedasticity-consistent, White) standard error:

$$\begin{aligned}\text{std. err}_{\text{rob}}(\hat\beta_{n,j}) &= \sqrt{[\hat V_n]_{jj}/n} \\ &= \sqrt{\Big[(X^\top X)^{-1}\Big(\sum_{i=1}^n \hat U_i^2 X_iX_i^\top\Big)(X^\top X)^{-1}\Big]_{jj}}\end{aligned}$$

- Recall: $\mathrm{E}\left[U \mid X\right]=0$, $\mathrm{Var}\left(U \mid X\right)=\mathrm{diag}(\sigma_1^2,\dots,\sigma_n^2)$:

$$\mathrm{Var}\left(\hat\beta \mid X\right) = (X^\top X)^{-1}\Big(\sum_{i=1}^n \sigma_i^2 X_iX_i^\top\Big)(X^\top X)^{-1}$$

- Robust estimator: $\hat U_i^2$ in place of σ_i^2.

t-ratio with robust standard errors

- Cramér convergence theorem, without proof: $Z_n \to_d Z$, $c_n \to_p c \neq 0$, c non-random:

$$\frac{Z_n}{c_n} \to_d \frac{Z}{c}$$

- $\sqrt{[\hat{V}_n]_{jj}} \to_p \sqrt{[V]_{jj}} > 0$; apply to the component CLT:

$$\frac{\hat{\beta}_{n,j} - \beta_j}{\text{std. err}_{\text{rob}}(\hat{\beta}_{n,j})} = \frac{\sqrt{n}(\hat{\beta}_{n,j} - \beta_j)}{\sqrt{[\hat{V}_n]_{jj}}} \to_d N(0,1)$$

- Asymptotic confidence interval:

$$CI_{1-\alpha} = \Big[\hat{\beta}_{n,j} - z_{1-\alpha/2}\,\text{std. err}_{\text{rob}}(\hat{\beta}_{n,j}), \\ \hat{\beta}_{n,j} + z_{1-\alpha/2}\,\text{std. err}_{\text{rob}}(\hat{\beta}_{n,j})\Big]$$

- Coverage, t-ratio $\to_d Z \sim N(0,1)$:

$$\begin{aligned} \text{P}(\beta_j \in CI_{1-\alpha}) &= \text{P}\Big(\Big|\frac{\hat{\beta}_{n,j} - \beta_j}{\text{std. err}_{\text{rob}}(\hat{\beta}_{n,j})}\Big| \le z_{1-\alpha/2}\Big) \\ &\to \text{P}(|Z| \le z_{1-\alpha/2}) \\ &= 1 - \alpha \end{aligned}$$

Usual standard errors

- Recall:

$$\text{std. err}(\hat{\beta}_{n,j}) = \sqrt{s^2\big[(X^\top X)^{-1}\big]_{jj}} = \sqrt{[s^2 \hat{Q}_n^{-1}]_{jj}/n}, \\ s^2 = \frac{1}{n-k}\sum_{i=1}^n \hat{U}_i^2$$

- Usual: $s^2\hat{Q}_n^{-1}$ estimates $\sigma^2 Q^{-1}$ under homoskedasticity.
- Robust: $\hat{V}_n$ estimates $V = Q^{-1}\Omega Q^{-1}$.

$s^2 \to_p \text{E}\left[U_i^2\right]$

- $\text{E}\left[U_i^2\right] < \infty$; weak LLN, consistency and Slutsky:

$$\begin{aligned} \frac{1}{n}\sum_{i=1}^n \hat{U}_i^2 = &\underbrace{\frac{1}{n}\sum_{i=1}^n U_i^2}_{\to_p \text{E}[U_i^2]} - 2\,\underbrace{(\hat{\beta}_n - \beta)^\top}_{\to_p 0^\top}\,\underbrace{\frac{1}{n}\sum_{i=1}^n X_i U_i}_{\to_p 0} \\ &+ \underbrace{(\hat{\beta}_n - \beta)^\top}_{\to_p 0^\top}\,\underbrace{\Big(\frac{1}{n}\sum_{i=1}^n X_i X_i^\top\Big)}_{\to_p Q}\,\underbrace{(\hat{\beta}_n - \beta)}_{\to_p 0} \\ \to_p &\ \text{E}\left[U_i^2\right] \end{aligned}$$

- Degrees-of-freedom correction, k fixed:

$$s^2 = \underbrace{\frac{n}{n-k}}_{\to 1}\,\underbrace{\frac{1}{n}\sum_{i=1}^n \hat{U}_i^2}_{\to_p \text{E}[U_i^2]} \to_p \text{E}\left[U_i^2\right]$$

- Homoskedasticity, law of iterated expectations:

$$\text{E}\left[U_i^2\right] = \text{E}\left[\text{E}\left[U_i^2 \mid X_i\right]\right] = \sigma^2$$

Homoskedasticity: robust and usual agree

- Both asymptotic-variance estimators consistent:

$$\widehat{V}_n \to_p \sigma^2 Q^{-1}, \qquad s^2 \widehat{Q}_n^{-1} \to_p \sigma^2 Q^{-1}$$

- Ratio of standard errors, $\sigma^2 [Q^{-1}]_{jj} > 0$; Slutsky:

$$\frac{\text{std. err}_{\text{rob}}(\hat{\beta}_{n,j})}{\text{std. err}(\hat{\beta}_{n,j})} = \sqrt{\frac{[\widehat{V}_n]_{jj}}{[s^2 \widehat{Q}_n^{-1}]_{jj}}} \to_p 1$$

- Usual t-ratio, $A_n Z_n \to_d AZ$:

$$\frac{\hat{\beta}_{n,j} - \beta_j}{\text{std. err}(\hat{\beta}_{n,j})} = \underbrace{\frac{\text{std. err}_{\text{rob}}(\hat{\beta}_{n,j})}{\text{std. err}(\hat{\beta}_{n,j})}}_{\to_p 1} \cdot \underbrace{\frac{\hat{\beta}_{n,j} - \beta_j}{\text{std. err}_{\text{rob}}(\hat{\beta}_{n,j})}}_{\to_d N(0,1)}$$
$$\to_d N(0,1)$$

Heteroskedasticity: usual standard errors fail

- Without homoskedasticity:

$$s^2 \widehat{Q}_n^{-1} \to_p \mathrm{E}\left[U_i^2\right] Q^{-1} \neq Q^{-1} \Omega Q^{-1} = V \quad \text{(in general)}$$

- Example, $k = 1$, no intercept: $Q = \mathrm{E}\left[X_i^2\right]$, $\Omega = \mathrm{E}\left[U_i^2 X_i^2\right]$:

$$V = \frac{\mathrm{E}\left[U_i^2 X_i^2\right]}{\left(\mathrm{E}\left[X_i^2\right]\right)^2}, \qquad \text{usual: } \frac{\mathrm{E}\left[U_i^2\right]}{\mathrm{E}\left[X_i^2\right]}$$
$$V - \frac{\mathrm{E}\left[U_i^2\right]}{\mathrm{E}\left[X_i^2\right]} = \frac{\mathrm{Cov}\left(U_i^2, X_i^2\right)}{\left(\mathrm{E}\left[X_i^2\right]\right)^2}$$

- $\mathrm{Cov}\left(U_i^2, X_i^2\right) > 0 \Longrightarrow$ usual standard error asymptotically too small.

Normality: exact and asymptotic

- Normal model: $U \mid X \sim N(0, \sigma^2 I_n)$; under Assumptions I and II:

$$\mathrm{Var}\left(\sqrt{n}(\hat{\beta}_n - \beta) \mid X\right) = \sigma^2 \widehat{Q}_n^{-1} \to_p \sigma^2 Q^{-1}$$

- σ^2 estimated by s^2:

$$\frac{\hat{\beta}_{n,j} - \beta_j}{\text{std. err}(\hat{\beta}_{n,j})} \Big| X \sim t_{n-k} \quad \text{for every } n > k$$
$$t_{n-k,\,1-\alpha/2} \to z_{1-\alpha/2}$$

- Normal model: $\mathrm{E}\left[U_i^2 \mid X_i\right] = \sigma^2$, $\mathrm{E}\left[U_i^4\right] = 3\sigma^4$.
- $\mathrm{E}\left[\|X_i\|^4\right] < \infty \Longrightarrow$ robust and usual standard-error ratio $\to_p 1$.

Usual and robust standard errors compared

- Validity of the t-ratio (large n: conditions for $\hat{V}_n \to_p V$):

Errors	Usual std. err	Robust std. err
$U \mid X \sim N(0, \sigma^2 I_n)$	exact, t_{n-k}	large n, $N(0,1)$
$\mathrm{E}\left[U_i^2 \mid X_i\right] = \sigma^2$	large n, $N(0,1)$	large n, $N(0,1)$
$\mathrm{E}\left[U_i^2 \mid X_i\right]$ depends on X_i	invalid in general	large n, $N(0,1)$

- Homoskedasticity: standard-error ratio $\to_p 1$, t quantile $\to z_{1-\alpha/2}$; asymptotically equivalent intervals:

$$\begin{aligned} \text{usual, exact (row 1):} \quad & \hat{\beta}_{n,j} \pm t_{n-k,\,1-\alpha/2}\,\text{std. err}(\hat{\beta}_{n,j}) \\ \text{usual, large } n\text{:} \quad & \hat{\beta}_{n,j} \pm z_{1-\alpha/2}\,\text{std. err}(\hat{\beta}_{n,j}) \\ \text{robust, large } n\text{:} \quad & \hat{\beta}_{n,j} \pm z_{1-\alpha/2}\,\text{std. err}_{\text{rob}}(\hat{\beta}_{n,j}) \end{aligned}$$

Summary

Consistency and asymptotic normality

- Assumptions for consistency $\Longrightarrow \hat{\beta}_n \to_p \beta$.
- Normalized representation:

$$\sqrt{n}(\hat{\beta}_n - \beta) = \Big(\frac{1}{n}\sum X_i X_i^\top\Big)^{-1} \frac{1}{\sqrt{n}} \sum X_i U_i$$

- Assumptions I to III:

$$\sqrt{n}(\hat{\beta}_n - \beta) \to_d N(0, V), \qquad V = Q^{-1}\Omega Q^{-1}$$

- $\mathrm{E}\left[U_i^2 \mid X_i\right] = \sigma^2 \Longrightarrow \Omega = \sigma^2 Q$, $V = \sigma^2 Q^{-1}$.

Variance estimation and standard errors

- $\hat{V}_n = \hat{Q}_n^{-1}\hat{\Omega}_n\hat{Q}_n^{-1}$, $\hat{Q}_n = \frac{1}{n}\sum X_i X_i^\top$, $\hat{\Omega}_n = \frac{1}{n}\sum \hat{U}_i^2 X_i X_i^\top$.
- Assumptions I to III, $\mathrm{E}\left[U_i^4\right] < \infty$, $\mathrm{E}\left[\|X_i\|^4\right] < \infty \Longrightarrow \hat{V}_n \to_p V$.
- std. $\text{err}_{\text{rob}}(\hat{\beta}_{n,j}) = \sqrt{[\hat{V}_n]_{jj}/n}$; robust t-ratio $\to_d N(0,1)$; normal confidence interval: coverage $\to 1-\alpha$.
- Usual: $s^2\hat{Q}_n^{-1} \to_p \mathrm{E}\left[U_i^2\right] Q^{-1}$; homoskedasticity: robust/usual standard-error ratio $\to_p 1$.
- Heteroskedasticity: usual standard errors invalid in general; robust valid under stated conditions.
- $U \mid X \sim N(0, \sigma^2 I_n)$: usual t-ratio exactly t_{n-k}; $t_{n-k,\,1-\alpha/2} \to z_{1-\alpha/2}$.